

Auteur: Noran Ben Said
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$$f(x, y) = (x - 3)^6 + (2x + y)^8$$

$$\frac{\partial f}{\partial x} = 6(x - 3)^5 + 16(2x + y)^7 = 0$$

$$\Leftrightarrow 6(x - 3)^5 = 0$$

$$\Rightarrow x = 3$$

$$\frac{\partial f}{\partial y} = 8(2x + y)^7 = 0$$

$$\Rightarrow y = -2x \Rightarrow y = -6$$

Punten: $(3, -6)$

$$H(x, y) = \begin{vmatrix} 30(x - 3)^4 + 224(2x + y)^6 & 112(2x + y)^6 \\ 224(2x + y)^6 & 112(2x + y)^6 \end{vmatrix}$$

$$\Rightarrow 30(x - 3)^4 + 224(2x + y)^6 \cdot 112(2x + y)^6 - 25088(2x + y)^{12}$$

$$H(3, -6) = 0 \Rightarrow ?$$

$$f(x, y) = (x - 3)^6 + (2x + y)^8 \geq 0$$

$$\left. \begin{array}{l} f(x, y) \geq 0 \\ f(3, -6) = 0 \end{array} \right\} \Rightarrow \text{minimum}$$

References